

Assignment-based Subjective Questions

1. From your analysis of the categorical variables from the dataset, what could you infer about their effect on the dependent variable?

Ans: Following inferences could be done:

- a) 'Season' - fall season had the most no. of people using boom bikes. Spring season had the least number.
- b) 'Yr' - 2019 attracted more number of booking from the previous year.
- c) 'Mnth' - Most of the bike bookings had been done during the month of May, June, July, Aug, Sep and Oct. This trend increased during starting of the year till mid of the year and then it started decreasing as we approached the end of year.
- d) 'Holiday' - Bikes were more used when there was no holiday.
- e) 'Weekday' - Sunday had the least users and the users increased from monday to friday.
- f) 'Workingday' - Booking seemed to be almost equal either on working day or non-working day.
- g) 'weathersit' - More people tend to use bikes when the weather was 'Clear. There was no case of bike booking in 'Heavy Rain' as usual.

2. Why is it important to use drop_first=True during dummy variable creation?

Ans: `drop_first = True` is important to use, because it helps in bringing down the extra column created during dummy variable creation. Hence it lessens the correlations created among dummy variables. We use `drop_first=True` to avoid dummy variable trap in machine learning. Dummy variable trap is when we have perfect multicollinearity between the dummy variables in linear model. If we didn't use this, then it can cause redundancy in regression model by violating the

assumptions of linear model.

- 3. Looking at the pair-plot among the numerical variables, which one has the highest correlation with the target variable?**

Ans: 'temp' variable has the highest correlation with the target variable.

- 4. How did you validate the assumptions of Linear Regression after building the model on the training set?**

Ans: Validation of all assumptions of MLR are done, after modelling the system.

- a) **Multicollinearity** - There should be insignificant multicollinearity among variables. This have been done initially through reducing the variables using RFE method and then discarding the variables on the basis of VIF score of remaining variables used. All variables should have VIF score less than 5.
- b) **Normality of error terms** - Error terms should be normally distributed. This is basically done through residual analysis(Histogram of residuals) (i.e plotting graph to show error are normally distributed with mean equal to zero).
- c) **Linear relationship** - Linearity should be visible among variables. This is basically done by calculating cost function or error function. Scatter plot and residual plot have been used to verify.

$$\text{cnt} = 0.150081 + \text{yr}*(0.232598) + \text{temp}*(0.502913) + \text{holiday}*(-0.100259) + \text{July}*(-0.052338) + \text{Sept}*(0.081182) + \text{Cloudy}*(-0.078458) + \text{Light_snow_rain}*(-0.298944) + \text{spring}*(-0.076589) + \text{summer}*(0.036651) + \text{winter}*(0.082879)$$

This equation shows that target variable 'cnt' depends positively high on 'temp'(i.e temperature) and 'yr'(i.e year) and negatively high on weather condition having Light_snow_rain.

- d) **Homoscedasticity** - There should be no visible pattern in residual values. This

is basically done by plotting graph between predicted target value and error value.(Residuals vs Fitted Plot).

- e) **Independence of residuals** - The residuals(errors) should be independent of each other (No auto-correlation).

5. Based on the final model, which are the top 3 features contributing significantly towards explaining the demand of the shared bikes?

Ans: Below are the top 3 features contributing significantly towards explaining the demand of the shared bikes :

- a) 'temp'
- b) 'yr'
- c) 'Light_snow_rain'

General Subjective Questions

1. Explain the linear regression algorithm in detail.

Ans: Linear regression may be defined as the statistical model that inspect the linear relationship between a dependent variable with given set of independent variables. Linear relationship between variables means that when the value of one or more independent variables will change (increase or decrease), the value of dependent variable will also change accordingly (increase or decrease).

Mathematically the relationship can be represented by the equation –

$$Y = mX + c$$

X - is the independent variable.

m - is the slope of the regression line.

c - is a constant, known as the Y-intercept.

It is one of the easiest and most popular Machine Learning algorithms. It is a statistical method that is used for predictive analysis. Linear regression makes predictions for continuous/real or numeric variables such as sales, salary, age, product price, etc.

Types of Linear Regression :

a) Simple Linear Regression:

If a single independent variable is used to predict the value of a numerical dependent variable, then such a Linear Regression algorithm is called Simple Linear Regression.

b) Multiple Linear regression:

If more than one independent variable is used to predict the value of a numerical dependent variable, then such a Linear Regression algorithm is called Multiple

Linear Regression.

Mathematically the relationship of multiple linear regression can be represented with the help of following equation –

$$Y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_i x_i + \varepsilon$$

Y= Dependent Variable.

X_i = Independent Variable.

β_0 = intercept of the line.

β_i = Linear regression coefficient.

ε = random error

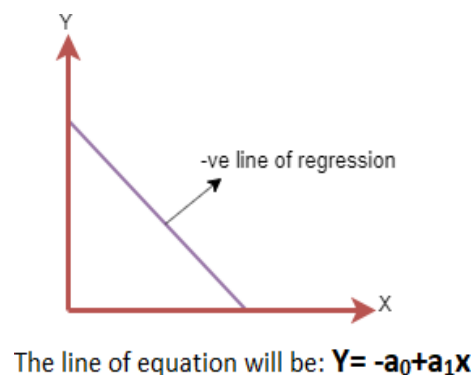
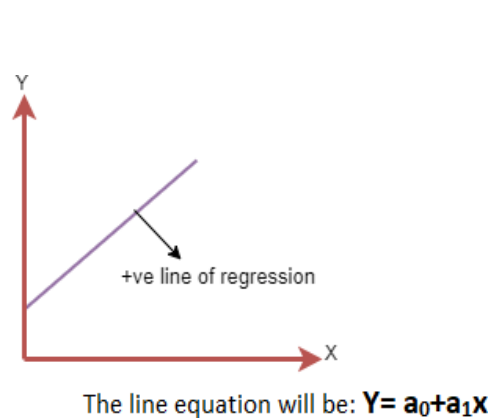
Linear Regression Line

1. Positive Linear Relationship:

If the dependent variable increases on the Y-axis and independent variable increases on X-axis, then such a relationship is termed as a Positive linear relationship.

2. Negative Linear Relationship:

If the dependent variable decreases on the Y-axis and independent variable increases on the X-axis, then such a relationship is called a negative linear relationship.



Assumptions -

The following are the assumptions that is made by Linear Regression model :

- a) **Multicollinearity** - There should be insignificant multicollinearity among variables. This have been done initially through reducing the variables using RFE method and then discarding the variables on the basis of VIF score of remaining variables used. All variables should have VIF score less than 5.
- b) **Normality of error terms** - Error terms should be normally distributed. This is basically done through residual analysis(Histogram of residuals) (i.e plotting graph to show error are normally distributed with mean equal to zero). Q-Q plot can also be done to check it.
- c) **Linear relationship** - Linearity should be visible among variables. This is basically done by calculating cost function or error function. Scatter plot and residual plot can be used to verify.
- d) **Homoscedasticity** - There should be no visible pattern in residual values. This is basically done by plotting graph between predicted target value and error value.(Residuals vs Fitted Plot). Breusch-Pagan or White's test can also be done to check it.
- f) **Independence of residuals** - The residuals(errors) should be independent of each other (No auto-correlation). Durbin-Watson test (closer to 2) can be used to check this assumption. Also for time series data Polt residuals over time.

2. Explain the Anscombe's quartet in detail.

Ans: Anscombe's Quartet is the famous example to demonstrate the importance of data visualization which was developed by the statistician Francis Anscombe in 1973 to signify both the importance of plotting data before analyzing it with statistical properties.It comprises of four data-set and each data-set consists of eleven (x,y) points. The basic thing to analyze about these data-sets is that they all share the same descriptive statistics(mean, variance, standard deviation etc) but

different graphical representation. Each graph plot shows the different behavior irrespective of statistical analysis.

x1	y1	x2	y2	x3	y3	x4	y4
10	8.04	10	9.14	10	7.46	8	6.58
8	6.95	8	8.14	8	6.77	8	5.76
13	7.58	13	8.74	13	12.74	8	7.71
9	8.81	9	8.77	9	7.11	8	8.84
11	8.33	11	9.26	11	7.81	8	8.47
14	9.96	14	8.1	14	8.84	8	7.04
6	7.24	6	6.13	6	6.08	8	5.25
4	4.26	4	3.1	4	5.39	19	12.5
12	10.84	12	9.13	12	8.15	8	5.56
7	4.82	7	7.26	7	6.42	8	7.91
5	5.68	5	4.74	5	5.73	8	6.89

Apply the statistical formula on the above data-set:

Average Value of $x = 9$

Average Value of $y =$

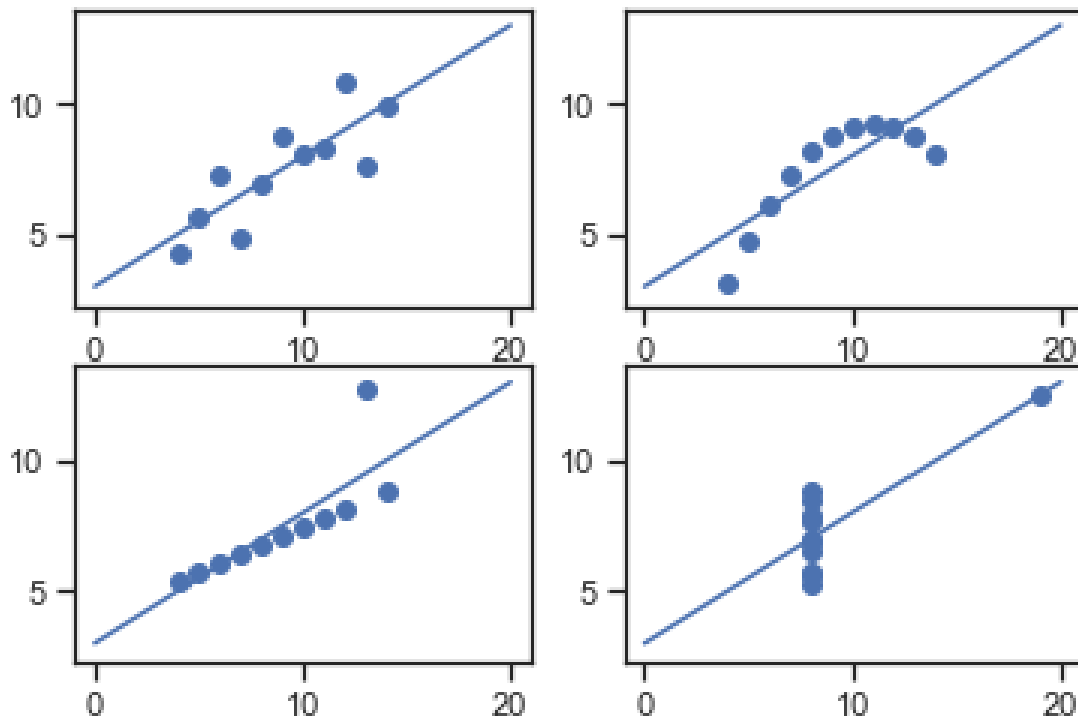
7.50 Variance of $x = 11$

Variance of $y = 4.12$

Correlation Coefficient = 0.816

Linear Regression Equation : $y = 0.5 x + 3$

However, the statistical analysis of these four data-sets are pretty much similar. But when we plot these four data-sets across the x & y coordinate plane, we get the following results & each pictorial view represent the different behavior.



Data-set I — consists of a set of (x,y) points that represent a linear relationship with some variance.

Data-set II — shows a curve shape but doesn't show a linear relationship (might be quadratic?).

Data-set III — looks like a tight linear relationship between x and y, except for one large outlier.

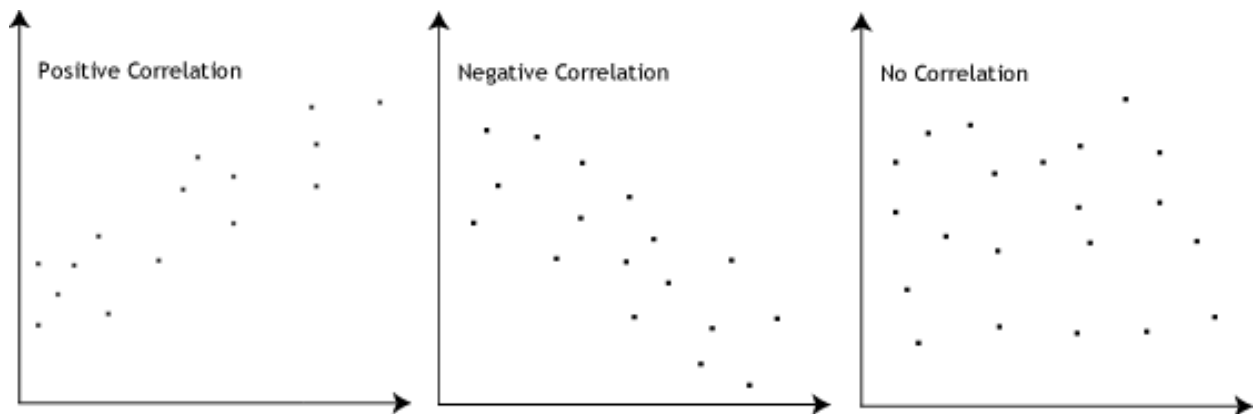
Data-set IV — looks like the value of x remains constant, except for one outlier as well.

Data-sets which are identical over a number of statistical properties, yet produce dissimilar graphs, are frequently used to illustrate the importance of graphical representations when exploring data.

3. What is Pearson's R?

Ans: Pearson's R, also known as the Pearson correlation coefficient, is a statistical measure that describes the linear relationship between two continuous variables. Its value ranges between -1 and 1, where -1 depicts a perfect negative linear relationship, 0 indicates no linear relationship, and 1 indicates a perfect positive linear relationship.

Pearson's R measures the degree to which the variables are related by calculating the ratio of the covariance between the variables to the product of their standard deviations. In other words, it measures how much the variables vary together relative to how much they vary independently.



Strength of Correlation:

- a) $|r| = 0.00 - 0.19$: Very weak correlation (the relationship between the variables is minimal).
- b) $|r| = 0.20 - 0.39$: Weak correlation (there's a small association between the variables).
- c) $|r| = 0.40 - 0.59$: Moderate correlation (a noticeable relationship exists).
- d) $|r| = 0.60 - 0.79$: Strong correlation (a clear association between the variables).
- e) $|r| = 0.80 - 1.00$: Very strong correlation (the variables are closely linked).

4. What is scaling? Why is scaling performed? What is the difference between normalized scaling and standardized scaling?

Ans: Scaling is a technique to standardize the independent features present in the data in a fixed range. It is performed during the data pre-processing to handle highly varying magnitudes or values or units. If feature scaling is not done, then a machine learning algorithm tends to weigh greater values, higher and consider smaller values as the lower values, regardless of the unit of the values.

Example: If an algorithm is not using feature scaling method then it can consider the value 300 cm to be greater than 2m but that's actually not true and in this case, the algorithm will give wrong predictions. So, we use Feature Scaling to bring all values to same magnitudes and thus, tackle this issue.

Normalization	Standardization
It is used when features are of different scales.	It is used when we want to ensure zero mean and unit standard deviation.
It is used when we want to ensure zero mean and unit standard deviation.	Mean and standard deviation is used for scaling.
Scales values between $[0, 1]$ or $[-1, 1]$.	It is not bounded to a certain range.
It is really affected by outliers.	It is much less affected by outliers.
It is often called as Scaling Normalization	It is often called as Z-Score Normalization.

5. You might have observed that sometimes the value of VIF is infinite. Why does this happen?

Ans: If there is perfect correlation, then $VIF = \text{infinity}$. A large value of VIF indicates that there is a correlation between the predictor. If the VIF is 6, this means that the variance of the model coefficient is inflated by a factor of 6 due to the presence of multicollinearity.

An infinite VIF value indicates that the corresponding predictor may be expressed exactly by a linear combination of other variables (which show an infinite VIF as well).

If there is perfect correlation, then $VIF = \text{infinity}$. This shows a perfect correlation between two independent variables. In the case of perfect correlation, we get $R^2 = 1$, which leads to $1/(1-R^2)$ infinity. To solve this problem we need to drop one of the variables from the dataset which is causing this perfect multicollinearity.

6. What is a Q-Q plot? Explain the use and importance of a Q-Q plot in linear regression.

Ans: The quantile-quantile (Q-Q) plot is a graphical technique for determining if two data sets come from populations with a common distribution.

Use of Q-Q plot:

A Q-Q plot is a plot of the quantiles of the first data set against the quantiles of the second dataset. By a quantile, we mean the fraction (or percent) of points below the given value. That is, the 0.3 (or 30%) quantile is the point at which 30% percent of the data fall below and 70% fall above that value. A 45-degree reference line is also plotted. If the two sets come from a population with the same distribution, the points should fall approximately along this reference line. The greater the departure from this reference line, the greater the evidence for the conclusion that the two data sets have come

from populations with different distributions.

Importance of Q-Q plot:

When there are two data samples, it is often desirable to know if the assumption of a common distribution is justified. If so, then location and scale estimators can pool both data sets to obtain estimates of the common location and scale. If two samples do differ, it is also useful to gain some understanding of the differences. The Q-Q plot can provide more insight into the nature of the difference than analytical methods such as the chi-square and Kolmogorov-Smirnov 2-sample tests.